

Movie Economics & Ratings Analytics

A TMDB × OMDb Data Analysis Pipeline

Top 200 Highest-Grossing Films – Exploratory Data Analysis Report

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“Note: TMDB budget/revenue and OMDb rating data are pulled live from public APIs. See Data Disclaimer (Section 15) for full context.”

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Section 1 – Introduction & Objective

Introduction

This report presents an end-to-end data analytics pipeline built on live movie data pulled from two public APIs: TMDB (The Movie Database) and OMDb (Open Movie Database). The pipeline fetches, cleans, verifies, joins, and analyzes the top 200 highest-grossing films on TMDB, producing a master dataset that combines box-office economics (budget, revenue, ROI) with critical and audience reception (IMDb rating, Rotten Tomatoes score, Metascore).

Objective

- Understand the distribution of production budgets, box-office revenue, ROI, and critical ratings across 200 high-grossing films.
- Analyze whether a bigger production budget is associated with better critical or audience reception.
- Measure how closely IMDb, Rotten Tomatoes, and Metascore agree with each other.
- Identify genre-level and studio-level patterns in output volume, ratings, and ROI.
- Segment films by budget bracket and uncover real-world film-economics patterns with data-backed recommendations.

Tools & Environment

Tool / Library	Purpose
Python 3	Core programming language
Requests + python-dotenv	Live TMDB & OMDb API connectivity
Pandas	Data manipulation & cleaning
Numpy	Numerical aggregation and correlation analysis
Matplotlib	Base plotting framework
Seaborn	Advanced statistical visualization
VS Code	Code editor & development environment

Section 2 – Dataset Overview

Movie metadata (budget, revenue, genres, runtime, popularity) was pulled live from TMDB's /discover/movie endpoint sorted by revenue (descending) across 10 pages, with a second per-movie call to /movie/{id} to retrieve budget, revenue, and imdb_id – fields the list endpoint does not return. Rating data (IMDb rating, Rotten Tomatoes score, Metascore, box office, awards) was pulled live from OMDb, queried once per imdb_id. The two tables were joined on imdb_id into a single master table.

Pipeline Summary

Metric	Value
Movies pulled (TMDB, revenue-sorted)	200
Movies cleaned & verified	200
Ratings matched via OMDb	200 / 200 (100%)
Final master table rows	200
Master table columns	25
Movies with a matched IMDb rating in the join	198 / 200 (99.0%)
Movies reporting a usable budget	199 / 200 (99.5%)
Movies reporting a usable revenue	200 / 200 (100.0%)

Column Description (Master Table)

Column	Nulls	Description
tmdb_id / imdb_id	0	Unique TMDB and IMDb identifiers used for the join
title / release_date / release_year	0	Film title and release date, with year extracted separately
genres / genre_primary	0	Full genre list (comma-joined) and the first-listed genre
budget / budget_reported	1	Production budget in USD; 0-from-API values are flagged and nulled, not treated as real zeros
revenue / revenue_reported	0	Box-office revenue in USD, same zero-flagging logic as budget
roi	1	(revenue - budget) / budget, set only where both are reported
runtime	0	Runtime in minutes (0 values nulled)
original_language	0	ISO language code of the original release
production_companies	0	Comma-joined list of production companies
popularity / tmdb_vote_average / tmdb_vote_count	0	TMDB's own popularity and audience-vote metrics
imdb_rating / imdb_votes	2 / 0	IMDb's audience rating (/10) and vote count
rotten_tomatoes	8	Rotten Tomatoes critic score (%), stripped from OMDb's string format
metascore	6	Metacritic aggregate score
rated / box_office / awards	4 / 5 / 2	MPAA rating, OMDb-reported box office, and awards text

Section 3 – Data Cleaning – Steps & Decisions

Every cleaning decision was made based on data evidence, not assumptions. Each step is documented with the reason behind it.

Step	Action	Reason
1	Drop rows with no imdb_id (movies)	Nothing to join to OMDb later without it
2	drop_duplicates() on tmdb_id	TMDB pagination can overlap across pages
3	release_date → datetime, extract release_year	Enables year-over-year trend analysis
4	Flatten genres / production_companies lists to strings	TMDB returns these as Python lists; genre_primary = first listed genre
5	budget = 0 / revenue = 0 → flagged then nulled	0 from TMDB means 'not reported', not 'the movie made \$0'
6	roi computed only where both are reported	Prevents a false ROI being computed against a null or fabricated budget
7	runtime = 0 → null	0-minute runtime is a missing value, not a real duration
8	drop_duplicates() on imdb_id (ratings)	One rating row expected per film
9	OMDb "N/A" string → proper null	OMDb returns the literal string "N/A" for missing numeric fields
10	imdb_votes: strip commas → pd.to_numeric()	Raw format was '1,486,308' as text
11	rotten_tomatoes: strip '%' → float	Raw format was '81%' as text
12	box_office: strip '\$' and commas → float	Raw format was '\$785,221,649' as text

Post-Cleaning Verification

Before the join, both cleaned tables were re-checked by an independent verification step. All checks passed on both tables:

- **Movies:** No duplicate tmdb_id, and no null imdb_id, in the movies table.
- **Movies:** release_year falls within a sane 1900–2027 range, and roi is only ever set where both budget_reported and revenue_reported are true.
- **Ratings:** No duplicate imdb_id, imdb_rating within 0–10, rotten_tomatoes within 0–100, and zero leftover "N/A" strings in the ratings table.

Join coverage: 100.0% of cleaned OMDb rows matched an existing movie record; 198 of 200 films (99.0%) ended up with a non-null IMDb rating after the left join.

Section 4 – Exploratory Data Analysis

Key Metrics Summary

Films Analyzed	Median Budget	Median Revenue	Avg IMDb Rating	Avg RT Score
200	\$160.0M	\$852.7M	7.20 / 10	74.1%

IMDb Rating Distribution (/10)

Metric	Value
Min	4.8
25th %ile	6.6
Median	7.3
Mean	7.20
75th %ile	7.8
Max	9.1
Std Dev	0.84

Rotten Tomatoes Distribution (%)

Metric	Value
Min	18
25th %ile	60.75
Median	79
Mean	74.08
75th %ile	91
Max	99
Std Dev	20.17

Budget Distribution (USD, reported only)

Metric	Value
Min	\$0.2M
25th %ile	\$111.5M
Median	\$160.0M
Mean	\$167.4M
75th %ile	\$200.0M
Max	\$658.8M
Std Dev	\$84.7M

Revenue Distribution (USD)

Metric	Value
Min	\$619.0M

Metric	Value
25th %ile	\$719.1M
Median	\$852.7M
Mean	\$954.9M
75th %ile	\$1,040.1M
Max	\$2,923.7M
Std Dev	\$376.9M

ROI Distribution – (revenue - budget) / budget

Metric	Value
Min	0.52x
25th %ile	3.20x
Median	4.53x
Mean	23.54x
75th %ile	7.49x
Max	3,349x
Std Dev	237.07x

Note: the mean ROI (23.5x) is far higher than the 75th percentile (7.5x) because ROI is heavily right-skewed – a handful of very low-budget breakout hits (indie animation, a first-mover blockbuster from decades ago) return ROI in the hundreds or thousands of multiples, pulling the mean far above what a typical film in this dataset actually returns. The median (4.53x) is the more representative figure for a ‘typical’ top-200 grosser.

Genre & Certification Snapshot

Category	Breakdown
Top genres by count	Action 60 (30.0%), Adventure 53 (26.5%), Animation 27 (13.5%), Science Fiction 17 (8.5%)
MPAA rating (‘rated’)	PG-13 116 (58%), PG 56 (28%), R 13 (6.5%), G 6 (3%), Not Rated 5 (2.5%)
Original language	English 189 (94.5%), Chinese 10 (5.0%), Japanese 1 (0.5%)
Runtime	Median 131 min, mean 130.3 min, range 86–201 min

Section 5 – Visualization 1 – Average ROI by Genre

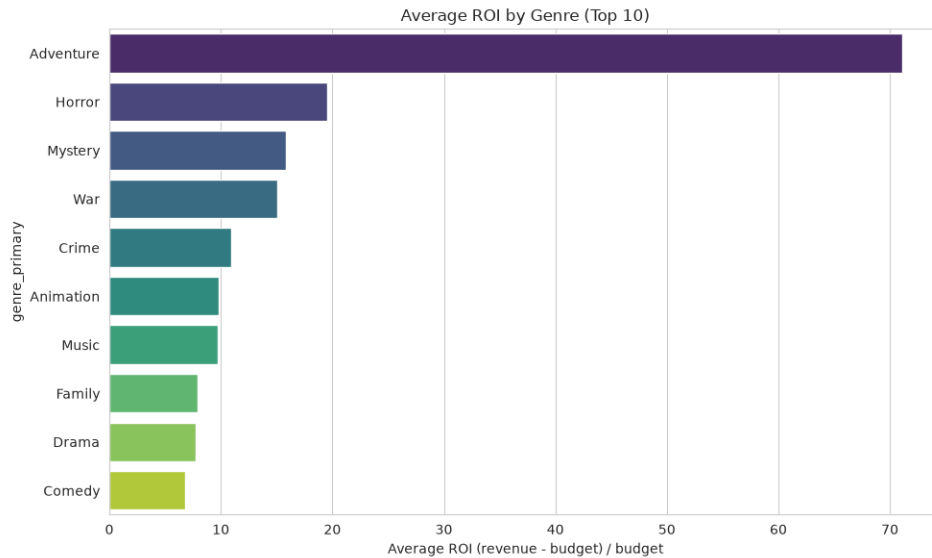


Fig 1. Average ROI (top 10 genres), computed only from films where both budget and revenue are reported.

What does this chart mean?

“Which kind of film gives you the most bang for your production dollar?”

Adventure sits far above every other genre, averaging roughly 71x ROI – driven by a small number of very low-budget titles (older classics and breakout animated hits) that went on to earn enormous multiples of their original budget. Horror, Mystery, and War form a distant second tier in the 15–20x range, while the highest-volume genres, Action and Adventure’s big-budget tentpoles, sit much closer to single-digit multiples once averaged across the whole genre.

Key Business Takeaways

- **Outlier-Driven Leader:** Adventure leads all genres with an average ROI of roughly 71x, but this figure is pulled upward by a handful of extreme outliers rather than being typical of the genre as a whole.
- **Lean Genres Punch Above Their Weight:** Horror, Mystery, and War post the next-strongest average multiples (15–20x), all genres associated with comparatively modest production budgets.
- **High-Volume Genres Return Less per Dollar:** Comedy, Drama, and Family cluster at the bottom of the top-10 (roughly 7–8x) – still profitable, but the least capital-efficient genres in this dataset.

Section 6 – Visualization 2 – IMDb Rating vs Rotten Tomatoes Score

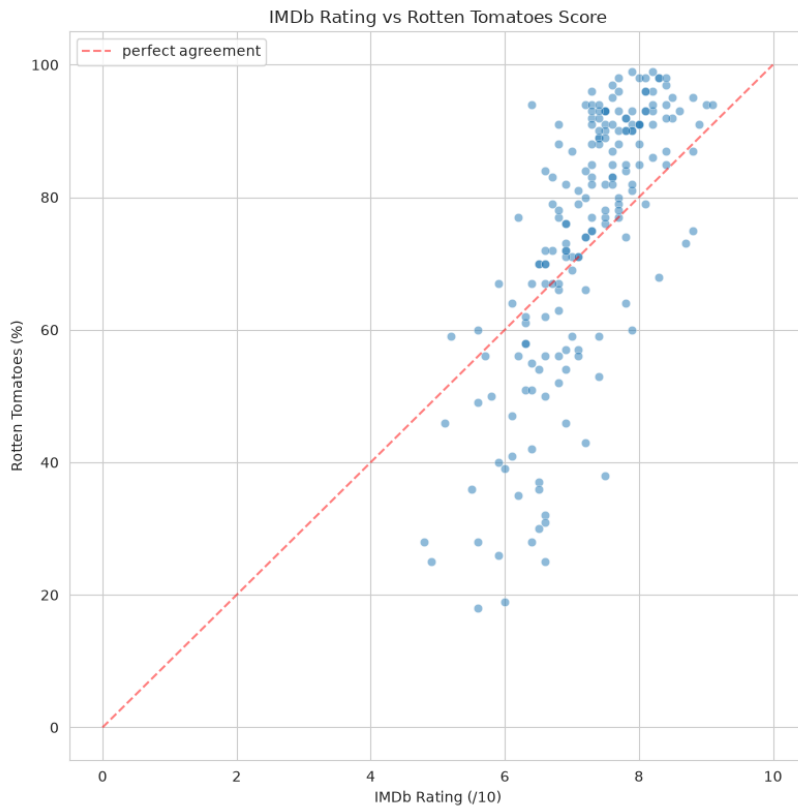


Fig 2. Scatter plot of IMDb rating (/10) against Rotten Tomatoes score (%), with a dashed reference line marking perfect agreement between the two scales.

What does this chart mean?

Each dot is one film, plotted by its audience score (IMDb, x-axis) against its critic score (Rotten Tomatoes, y-axis). Points sitting exactly on the dashed red line would mean the two scales agree perfectly. Points above the line mean critics rated the film relatively higher than audiences; points below mean audiences rated it relatively higher than critics.

Key Business Takeaways

- **Broad Agreement:** Across 192 films with both scores available, IMDb and Rotten Tomatoes correlate at 0.75 – a fairly strong, reasonably aligned relationship overall.
- **A Mild Critic-Friendly Skew:** Most points sit slightly above the perfect-agreement line, meaning Rotten Tomatoes' critic score tends to run a little more generous, in percentage terms, than the equivalent IMDb audience rating.
- **Notable Divergence Cases Exist:** A visible cluster of films sits well below the line – solid IMDb scores paired with much weaker Rotten Tomatoes scores. 'Transformers: Revenge of the Fallen' is the most extreme case in the dataset: IMDb 6.0/10 vs Rotten Tomatoes 19%.

Section 7 – Visualization 3 – Budget vs Revenue

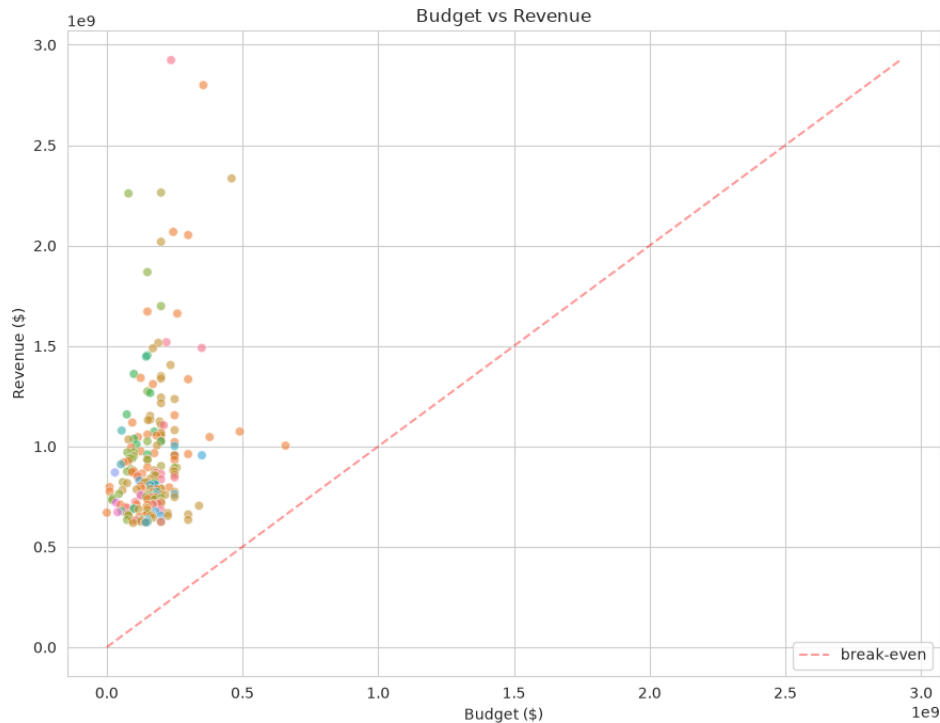


Fig 3. Every film plotted by production budget (x-axis) against box-office revenue (y-axis), colored by primary genre. The dashed line marks break-even (revenue = budget).

What does this chart mean?

Because this dataset is TMDB’s top 200 films sorted by revenue, every single film sits well above the break-even line – that is the nature of a highest-grossing list, not evidence that all films are profitable. What the chart actually reveals is that budget size explains very little of where a film lands on the revenue axis: some of the very highest revenue outcomes come from mid-sized budgets, while several of the largest budgets in the dataset produced comparatively modest revenue relative to their spend.

Key Business Takeaways

- **Selection Effect:** Because this is a highest-grossing-films dataset by construction, every point sits above break-even – this chart should not be read as ‘most movies are profitable’, but as how efficiently different budget levels convert into revenue within an already-successful slate.
- **Mid-High Budgets Punch Above Their Weight:** Several of the highest revenue outcomes in the dataset (above \$2B) come from budgets in the \$200–350M range rather than the very largest budgets, showing that scale alone does not guarantee the top revenue tier.
- **Budget Is a Weak Revenue Predictor:** Correlation between budget and revenue is a moderate 0.31 – a real but far from deterministic relationship.

Section 8 – Visualization 4 – Movie Count by Genre

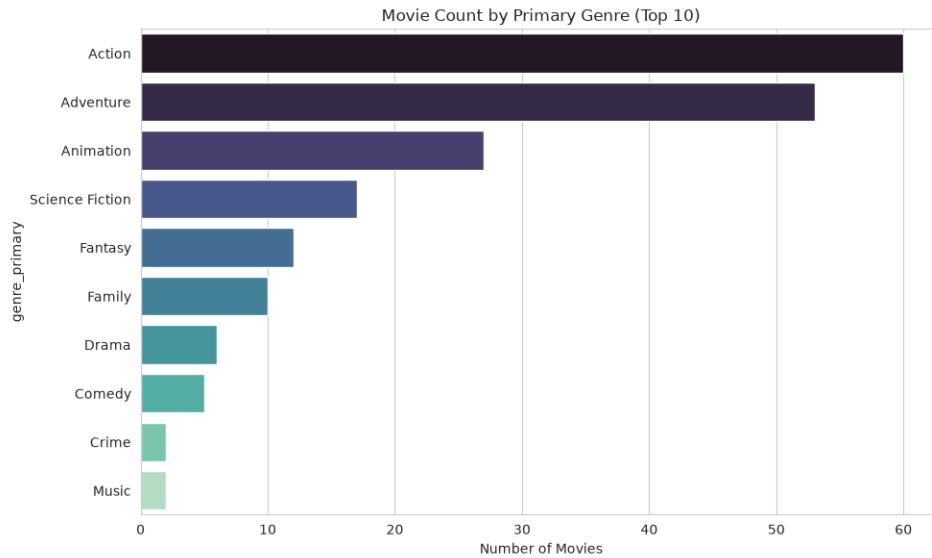


Fig 4. Top 10 primary genres by number of films in the top-200 grossing dataset.

What does this chart mean?

This chart breaks down the top-200 highest-grossing films by primary genre. Action and Adventure dominate by a wide margin, together accounting for well over half the dataset, while Animation forms a solid third tier.

Key Business Takeaways

- **Action & Adventure Dominate:** Action (60 films) and Adventure (53 films) together make up 56.5% of the entire top-200 list – the highest-grossing space is heavily concentrated in these two genres.
- **Animation Is the Strong Third Tier:** Animation (27 films) is the clear third genre, and, as later sections show, carries the highest average IMDb rating among the high-volume genres.
- **Prestige & Niche Genres Are Underrepresented:** Genres like Drama (6), Comedy (5), Crime (2), and Horror (1) appear rarely in the highest-grossing bracket – they exist in the broader film market but are simply not where the biggest box-office numbers concentrate.

Section 9 – Visualization 5 – Average IMDb Rating by Release Year

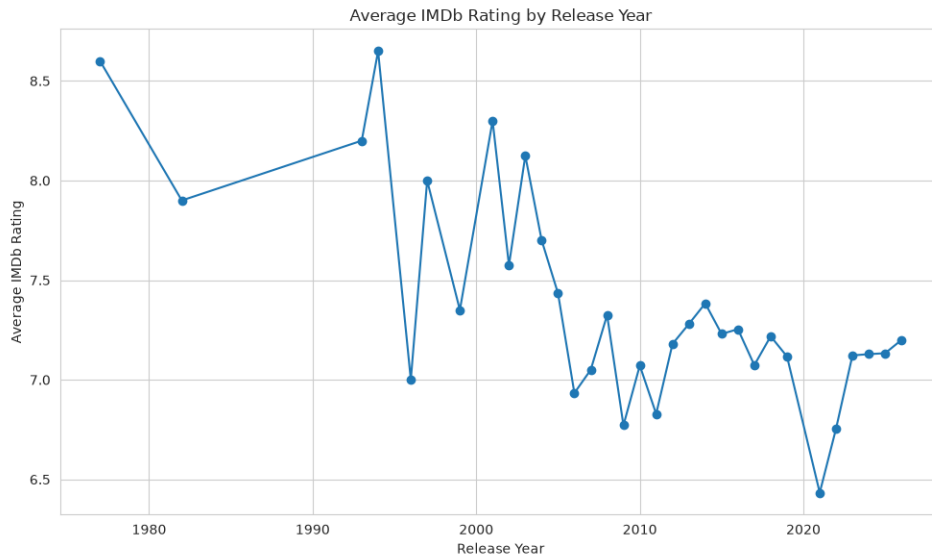


Fig 5. Average IMDb rating of top-200 grossing films, by release year (1977–2026).

What does this chart mean?

Each point is the average IMDb rating of the top-grossing films released in that year. The earliest years (late 1970s to mid-1990s) sit noticeably higher, before a clear step down starting in the mid-1990s and a broadly flatter, lower band from the 2000s onward.

Key Business Takeaways

- **A Clear Long-Term Decline:** From 1977 to 2026, the average IMDb rating of top-grossing films fell from 8.60 to 7.20 – a meaningful long-run decline.
- **Early Years Are Thin Samples:** The very earliest years in the dataset (1977, 1994) are represented by only one or two iconic titles each, so their high averages reflect a small, hand-picked sample rather than an entire year's slate.
- **Recent Years Have Stabilized:** From roughly 2010 onward, the yearly average settles into a comparatively narrow 6.4–7.4 band, suggesting the modern high-grossing slate is more consistent, if lower on average, than the earlier era of the dataset.

Section 10 – Genre & Studio Analysis

Beyond raw counts, it is worth looking at which genres and studios actually deliver the strongest ratings, and which production companies dominate the highest-grossing bracket by volume.

Average Rating by Genre (High-Volume Genres)

Genre	Films	Avg IMDb Rating	Avg Rotten Tomatoes
Animation	27	7.47	82.4%
Adventure	53	7.25	72.1%
Science Fiction	17	7.24	72.0%
Drama	6	7.20	76.3%
Action	60	7.17	75.7%
Comedy	5	7.06	75.8%
Fantasy	12	6.88	63.8%
Family	10	6.83	72.4%

Top Studios by Volume (Top-200 Grossing List)

Studio (Primary)	Films in Top 200
Marvel Studios	24
Warner Bros. Pictures	15
Walt Disney Pictures	14
Pixar	13
Lucasfilm Ltd.	8
Universal Pictures	8
DreamWorks Animation	8
Paramount Pictures	7
Walt Disney Animation Studios	7
New Line Cinema	7

Marvel Studios, Warner Bros., Disney, and Pixar together account for 66 of the 200 films in this list (33.0%) – roughly one in three of the highest-grossing films in the dataset comes from just four studios, reflecting how concentrated franchise and IP-driven filmmaking is at the top of the box office.

Section 11 – Budget Bracket Analysis

The 199 films with a reported budget were segmented into three brackets to understand how spending level relates to both financial return and critical reception.

Bracket	Budget Range	Films	Share	Avg ROI	Avg IMDb Rating
Lean	Under \$100M	38	19.1%	104.2x	7.42
Mid-Budget	\$100M – \$200M	93	46.7%	5.15x	7.21
Tentpole	Above \$200M	68	34.2%	3.59x	7.09

The Lean bracket's 104.2x average ROI is driven almost entirely by a small number of extreme outliers (a handful of sub-\$20M films that became massive hits), not by lean films being reliably profitable as a rule. The Mid-Budget bracket is both the largest segment (46.7% of films) and delivers a strong, more consistent average return (5.15x) alongside the highest average IMDb rating outside the Lean tier.

Observations

- **The Efficiency Sweet Spot:** Even after discounting the outlier-driven Lean-bracket average, the Mid-Budget tier (\$100–200M) offers the best combination of scale, consistency, and return in the dataset.
- **Bigger Budgets Trend Toward Slightly Lower Ratings:** Average IMDb rating declines steadily as budget bracket increases (7.42 → 7.21 → 7.09) – bigger spending is mildly associated with lower, not higher, audience ratings in this dataset.
- **Tentpoles Still Work, Just Less Efficiently:** Tentpole films (above \$200M) still deliver solid absolute revenue and remain the smallest-ROI-multiple bracket, consistent with the diminishing-returns pattern typical of blockbuster-scale filmmaking.

Section 12 – Real-World Problems & Solutions

Based on the data patterns analyzed above, studios and production decision-makers face specific, data-backed challenges. Here are the core problems and proposed solutions.

Problem 1: Budget Does Not Buy Better Reception

- **The Data:** Budget correlates with IMDb rating at -0.13, with Rotten Tomatoes at -0.20, and with Metascore at -0.15 – all weak, and all slightly negative.
- **The Solution:** Marginal production spend beyond a certain point should not be justified on the assumption that it will improve critical or audience reception. Investment in story, casting, and execution matters more than raw budget scale for rating outcomes.

Problem 2: Diminishing ROI Efficiency at the Top of the Budget Ladder

- **The Data:** Average ROI falls steadily from the Lean bracket to the Tentpole bracket, while the mid-budget tier (\$100–200M) captures nearly half the entire top-200 list with a strong, more repeatable 5.15x average return.
- **The Solution:** A portfolio approach that balances a smaller number of tentpole bets with a larger base of mid-budget films is likely to be more capital-efficient than concentrating spend at the very top of the budget range.

Problem 3: Single-Source Rating Risk

- **The Data:** While IMDb and Rotten Tomatoes correlate at a fairly strong 0.75 overall, individual titles can diverge sharply – ‘Transformers: Revenge of the Fallen’ shows a 6.0/10 IMDb rating against just a 19% Rotten Tomatoes score.
- **The Solution:** Greenlighting sequels, franchise extensions, or marketing spend based on a single rating source risks acting on a misleading signal. Blending audience (IMDb) and critic (Rotten Tomatoes, Metascore) signals gives a more complete read on reception.

Section 13 – Strategic Recommendations

- **Prioritize the Mid-Budget Tier:** Mid-budget films (\$100–200M) deliver the strongest combination of volume (46.7% of the top-200 list), consistency, and average ROI (5.15x) – this is the most defensible tier for sustained investment.
- **Do Not Equate Budget With Quality:** Since budget correlates weakly, and slightly negatively, with all three rating measures, do not treat a larger budget as a proxy for a better-received film when allocating marketing or awards-campaign resources.
- **Track Multiple Rating Sources:** Because IMDb, Rotten Tomatoes, and Metascore only pairwise-correlate in the 0.66–0.89 range, use a blended score across all three rather than any single source for greenlight, marketing, or franchise-continuation decisions.
- **Lean Into Animation’s Rating Advantage:** Animation posts the highest average IMDb rating (7.47) among all high-volume genres, while Action and Adventure together already account for 56.5% of the highest-grossing slate and are highly saturated – Animation represents a comparatively under-leveraged quality signal.
- **Recognize Studio Concentration:** Marvel Studios, Warner Bros., Disney, and Pixar already account for a third of the top-200 list – new entrants competing purely on IP scale face a steep concentration advantage held by a small number of major studios.

Section 14 – Technical Challenges & Problem Solving

Real-world API data is rarely clean or complete. Executing this pipeline required overcoming specific technical constraints.

Dual-API Join Complexity

TMDB's /discover/movie endpoint does not return budget, revenue, or imdb_id – only its /movie/{id} details endpoint does. Solved by making a second API call per movie inside fetch.py, so every movie in the raw pull already carries the fields the rest of the pipeline depends on.

Rate-Limited Free-Tier APIs

Both TMDB and OMDb free tiers are rate-limited. Small delays were added between calls in fetch.py so a full run does not get throttled or blocked partway through.

Inconsistent Missing-Value Encoding

OMDb returns the literal string “N/A” for any field it does not have, rather than a proper null. Solved with an explicit na_like check in cleaning.py before any numeric conversion is attempted, so “N/A” never silently becomes a bad numeric value.

False Zeros in Budget & Revenue

TMDB returns 0 for a budget or revenue it simply does not have on file – not for a movie that actually made \$0. Solved by flagging budget_reported and revenue_reported booleans before nulling the zero values, so ROI and every downstream aggregate never silently treat missing data as a real zero.

Post-Cleaning Verification Layer

verification.py independently re-checks each cleaned table (uniqueness, valid value ranges, and confirming no leftover “N/A” strings) before the join into master_table.csv is allowed to run, catching any upstream cleaning bugs before they can reach the final dataset.

Section 15 – Conclusion

This analysis of the top 200 highest-grossing films successfully joins live TMDB budget/revenue data with live OMDb rating data into a single, verified master dataset, transforming two separate, messy API responses into actionable film-economics intelligence. The findings show that within this highest-grossing slate, budget scale is a weak predictor of both revenue outcome and critical reception, while the mid-budget tier and a small number of dominant studios and genres (Animation in particular) carry the strongest quality signals.

By executing this entire project, from live dual-API data collection and cleaning through independent verification, joining, and statistical visualization, this report demonstrates end-to-end analytical capability across the full data pipeline, from raw API response to strategic recommendation.

Data Disclaimer & Real-World Constraints

In real-world data science, data pulled from third-party APIs is never perfectly complete. TMDB and OMDb each have their own reporting gaps, format inconsistencies, and coverage limits, and this pipeline preserves that reality rather than hiding it: budget and revenue ‘not reported’ cases are explicitly flagged rather than treated as zero, and 2 of the 200 films still carry no matched IMDb rating after the join.

Because the dataset is, by construction, TMDB’s top 200 films sorted by revenue, every insight here describes patterns within an already highly successful slate of films – it does not describe the broader film market, where most productions do not reach this level of box-office performance. Correlation figures throughout this report (for example, budget vs. rating, or IMDb vs. Rotten Tomatoes) describe association, not causation, and should be read as directional, data-driven guidance rather than deterministic rules.

Prepared by: Viraj More | July 2026